MarketSight

Stock Market Prediction System

User's Manual

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1. Introduction

What is MarketSight?

MarketSight is a web-based stock market prediction platform that uses machine learning to forecast stock prices. The system analyzes historical market data and calculates technical indicators to generate predictions with confidence scores, helping you make more informed investment decisions.

Who is this manual for?

This manual is written for **end users** of the MarketSight platform. No programming or technical background is required. Whether you are a retail investor, a finance student, or simply curious about stock market predictions, this guide will walk you through every feature of the application.

Key Features at a Glance

Feature	Description
Stock Browser	Search and browse stocks by name, symbol, or sector
Interactive Charts	View candlestick and line charts with adjustable time ranges
ML Predictions	Get AI-powered price forecasts for individual stocks or all at once
Confidence Scores	See how confident the model is in each prediction (color-coded)
Prediction Overlay	Visualize the predicted price directly on the stock chart
User Settings	Manage your profile, change your password, and control your account
Admin Panel	Manage user accounts (superusers only)
Dark Mode	Switch between light, dark, and system-matched themes

2. Getting Started

Accessing MarketSight

Open your web browser and navigate to the MarketSight URL provided by your administrator. In a local development environment, this is typically:

```
http://localhost:5173
```

The Welcome Page

When you first visit MarketSight, you will see the **Welcome Page**. This page provides a brief overview of the platform's capabilities:

- **Historical Data** - Explore historical price data across multiple stocks
- **AI Predictions** - View data-driven forecasts powered by advanced ML models
- **Flexible Filtering** - Filter by date, exchange, and asset type

To begin using the application, click the **"Start exploring"** button. This will take you to the main Dashboard.

Navigation

MarketSight uses a **sidebar navigation** on the left side of the screen. The sidebar contains links to all major sections:

Menu Item	Icon	Description
Dashboard	Home	Your main overview page with summary statistics
Stocks	Chart	Browse, search, and view stock data and charts
Predictions	(via Stocks page)	Generate ML-powered price predictions
Settings	Gear	Manage your profile and account preferences
Admin	Users	Manage user accounts (visible to superusers only)

You can **collapse the sidebar** by clicking the menu button in the top-left corner. This gives you more screen space for charts and data. Click it again to expand.

At the bottom of the sidebar, you will find:

- **Theme switcher** - Change between light, dark, and system themes
- **User menu** - View your profile info and log out

3. Dashboard Overview

The Dashboard is your home screen. It gives you a quick snapshot of the system's current state through **four summary cards**:

Summary Cards

Card	What It Shows
Total Stocks	The number of stock symbols currently being tracked by MarketSight
Sectors	The number of unique market sectors represented (e.g., Technology, Healthcare)
Model Version	The current version of the ML model generating predictions
API Status	Whether the backend system is online and operational (green = healthy)

These cards update automatically when the page loads. If the API Status shows anything other than "Online", the system may be experiencing issues - see the Troubleshooting section.

4. Browsing Stocks

Navigate to the **Stocks** page by clicking "Stocks" in the sidebar.

Searching for Stocks

At the top of the page, you will find a **search box**. Type any part of a stock's symbol or company name to instantly filter the list. For example:

- Typing `AAPL` will show Apple Inc.
- Typing `Tech` will show all stocks in technology-related sectors

The results update as you type - no need to press Enter.

Viewing Modes

The Stocks page has two tabs:

All Stocks (default)

- Displays every stock in a single table
- Columns: Symbol, Name, Sector, Exchange
- Click any stock's **symbol** (shown in blue) to view its detail page

By Sector

- Groups stocks by their market sector (e.g., Technology, Healthcare, Energy)
- Each sector appears as a collapsible section
- Useful for comparing stocks within the same industry

Stock Table Columns

Column	Description
Symbol	The stock's ticker symbol (e.g., AAPL, MSFT). Click to view details.
Name	The full company name
Sector	The market sector the company belongs to. Shows "N/A" if not classified.

Exchange	The stock exchange where the stock is listed (e.g., NASDAQ, NYSE)

> **Tip:** If you see "No stocks available", the market data has not been loaded yet. Contact your administrator.

5. Stock Details & Charts

Click on any stock symbol from the Stocks page to open its **detail page**. This is where you can analyze historical price data and view ML predictions.

Stock Header

At the top of the page, you will see:

- The **company name** and **ticker symbol**
- Badges showing the stock's **sector**, **industry**, and **exchange**
- A **back arrow** button to return to the Stocks list

The Chart Tab

The Chart tab is selected by default and is the main feature of the stock detail page.

Price Information Bar

Above the chart, a row of key metrics is displayed:

Metric	Description
Current Price	The most recent closing price in dollars
Price Change	How much the price changed from the previous day (in dollars and percent). Green = up, Red = down.
Latest Volume	The number of shares traded in the most recent session

When the prediction overlay is enabled, additional metrics appear:

Metric	Description
Predicted Price (60d)	The model's forecasted price. Green if higher than current, red if lower.
Predicted Return	The expected percentage change from current price to predicted price
Confidence	How confident the model is in this prediction (see Section 7)

Chart Types

Two chart types are available. Toggle between them using the buttons in the top-right corner of the chart:

Candlestick Chart

- Each candle represents one trading day
- **Green candle** = the stock closed higher than it opened (bullish day)
- **Red candle** = the stock closed lower than it opened (bearish day)
- The thin lines (wicks) show the day's high and low prices
- Best for: Analyzing daily price action and trends

Line Chart

- Shows the closing price as a continuous blue line
- Simpler and cleaner view of the overall price trend
- Best for: Getting a quick sense of the stock's direction

Time Range Selection

Filter how much historical data is shown on the chart:

Button	Range	Best For
1D	Last 1 day	Intraday detail
1W	Last 7 days	Short-term trend
1M	Last 30 days	Monthly overview
ALL	All available data	Long-term analysis

Prediction Overlay

To see the ML prediction visualized on the chart:

1. Find the "**Show Prediction**" toggle switch in the control panel
2. Flip the switch to **ON**
3. The system will generate a prediction (you may see "Loading..." briefly)
4. An **orange dashed line** will appear on the chart, extending from the last known price to the predicted future price
5. The chart automatically zooms to the **1W** time range for the best view

> **Tip:** The prediction overlay is a great way to visually compare where the model thinks the price is headed versus recent price action.

The Details Tab

Switch to the Details tab to see the stock's metadata in a card layout:

- Symbol, Name, Sector, Industry, and Exchange

6. Getting Predictions

The **Predictions** page is your hub for generating and viewing ML-powered stock price forecasts. Navigate to it by clicking "Predictions" in the sidebar or through the Stocks page.

Generating a Single Prediction

1. On the Predictions page, find the **"Generate Prediction"** card
2. Click the **stock selector dropdown** and choose a stock (e.g., "AAPL - Apple Inc.")
3. Click the **"Predict"** button
4. Wait a moment while the model generates the forecast (the button will show a loading spinner)
5. A **prediction card** will appear below with the results

Predicting All Stocks at Once

If you want forecasts for every tracked stock:

1. Click the **"Predict All"** button (to the right of the Predict button)
2. The system will generate predictions for all active stocks
3. Multiple prediction cards will appear in a grid layout

Reading a Prediction Card

Each prediction card contains:

Field	Description
Symbol	The stock ticker (large, bold, top-left)
Return Badge	A colored badge showing the predicted return. Green with an up arrow for positive returns, red with a down arrow for negative returns.
Prediction Date	When this prediction was generated
Current Price	The stock's most recent closing price
Predicted Price	The model's forecasted price. Green text = price expected to go up, red = expected to go down.

Confidence	How reliable the model considers this prediction (shown as a percentage)
Model Version	Which version of the ML model was used
View Chart	Button to jump directly to the stock's chart with the prediction overlay

What to Do With Predictions

- **Compare** the predicted price against the current price to gauge expected movement
- **Check the confidence score** before acting on any prediction (see next section)
- **Click "View Chart"** to see the prediction in the context of recent price history
- **Use "Predict All"** to quickly scan for the strongest predicted movers across your tracked stocks

> **Note:** Predictions are forecasts, not guarantees. Always use predictions as one input among many in your decision-making process.

7. Understanding Confidence Scores

Every prediction comes with a **confidence score** that tells you how reliable the model considers its forecast. This score ranges from 0% to 100%.

How Confidence is Calculated

The confidence score is based on four market factors:

Factor	What It Measures
Volatility	How much the stock's price has been fluctuating recently. Lower volatility = higher confidence.
RSI (Relative Strength Index)	Whether the stock is overbought or oversold. Extreme RSI values reduce confidence.
Volume	How actively the stock is being traded. Higher volume generally means more reliable predictions.
Return Magnitude	How large the predicted price change is. Very large predicted moves have lower confidence.

Color Coding

Confidence is displayed with intuitive color coding throughout the application:

Color	Range	Meaning
Green	Above 70%	**High confidence.** The model is relatively sure about this prediction. Market conditions are stable and the predicted move is within normal range.
Yellow	40% - 70%	**Moderate confidence.** Some uncertainty exists. The stock may be experiencing higher volatility or unusual trading patterns.
Red	Below 40%	**Low confidence.** The model is uncertain. This could be due to extreme market conditions, very low volume, or a large predicted move. Treat with caution.

How to Use Confidence Scores

- **High confidence (green):** The prediction is based on stable conditions. Consider it a stronger signal.
- **Moderate confidence (yellow):** Useful as a directional indicator, but verify with other analysis.
- **Low confidence (red):** The model is uncertain. Do not rely heavily on this prediction. Investigate why conditions may be unusual.

> **Important:** A high confidence score does NOT guarantee accuracy. It means the model's input conditions are favorable for making a prediction. Always combine ML predictions with your own research and analysis.

8. Account Settings

Navigate to **Settings** in the sidebar to manage your account. The Settings page has three tabs.

My Profile

View and edit your personal information.

To view your profile:

- Your full name and email are displayed on the page

To edit your profile:

1. Click the **"Edit"** button
2. Modify your **Full Name** or **Email** in the input fields
3. Click **"Save"** to apply changes, or **"Cancel"** to discard them

> The Save button is only clickable when you have actually changed something.

Change Password

Update your account password.

1. Switch to the **Password** tab
2. Enter your **Current Password**
3. Enter your **New Password** (must be at least 8 characters)
4. Enter the new password again in the **Confirm Password** field
5. Click **"Update Password"**

Each password field has an **eye icon** you can click to temporarily show the password as you type.

> **Tip:** If you forget your current password, contact your administrator to reset it.

Delete Account (Danger Zone)

Permanently remove your account and all associated data.

1. Switch to the **Danger Zone** tab
2. Click the red "**Delete Account**" button
3. A confirmation dialog will appear warning that all data will be permanently deleted
4. Click "**Confirm Delete**" to proceed, or "**Cancel**" to go back

> **Warning:** This action cannot be undone. All your account data will be permanently deleted.

9. Admin Panel (Superusers Only)

The Admin page is only visible to users with **superuser** privileges. It allows you to manage all user accounts in the system.

Navigate to **Admin** in the sidebar.

Viewing Users

The Admin page displays a table of all registered users:

Column	Description
Full Name	The user's display name. Your own row is marked with a "You" badge.
Email	The user's email address
Role	Either "Superuser" (full admin access) or "User" (standard access)
Status	Green dot = Active, Gray dot = Inactive
Actions	Menu button for editing or deleting the user

Adding a New User

1. Click the **"Add User"** button (top-right, with a + icon)
2. Fill in the required fields:
   - **Email** (required) - must be a valid email address
   - **Full Name** (optional)
   - **Password** (required) - at least 8 characters
   - **Confirm Password** (required) - must match
3. Optionally check:
   - **Is superuser?** - gives the user admin access
   - **Is active?** - whether the account is enabled
4. Click **"Save"**

Editing a User

1. Click the **three-dot menu** on the user's row

2. Select **"Edit User"**
3. Modify any fields as needed (password fields are optional - leave blank to keep the current password)
4. Click **"Save"**

Deleting a User

1. Click the **three-dot menu** on the user's row
2. Select **"Delete User"**
3. A confirmation dialog will warn you that all associated data will be permanently deleted
4. Click **"Delete"** to confirm, or **"Cancel"** to go back

> **Warning:** Deleting a user is permanent and cannot be undone.

10. Appearance & Themes

MarketSight supports three appearance modes so you can use the application comfortably in any lighting condition.

Changing the Theme

1. Look at the **bottom of the sidebar**
2. Click the **theme switcher** dropdown
3. Select one of three options:

Theme	Description
Light	White backgrounds with dark text. Best for well-lit environments.
Dark	Dark backgrounds with light text. Easier on the eyes in low-light conditions.
System	Automatically matches your operating system's theme setting. If your OS switches between light and dark mode, MarketSight will follow.

Your theme preference is saved and will persist across browser sessions.

11. Troubleshooting & FAQ

Common Issues

Q: The dashboard shows "Loading..." and never finishes.

> The backend server may be down. Check that the API Status card shows "Online". If it doesn't, contact your system administrator to restart the backend services.

Q: I see "No stocks available" on the Stocks page.

> Market data has not been loaded into the system yet. An administrator needs to run the data seeding process, or the Airflow pipeline needs to complete its first run.

Q: My prediction failed with an error message.

> This can happen if:
>
> - The stock doesn't have enough historical data (at least 30 days of OHLC data is needed)
> - The ML model is not loaded on the backend
> - The backend server is temporarily unavailable
>
> Try again in a few minutes. If the problem persists, contact your administrator.

Q: The prediction overlay isn't showing on the chart.

> Make sure the "Show Prediction" toggle is switched ON. If it shows "Loading..." for an extended time, the prediction may have failed silently. Try toggling it off and on again.

Q: I can't see the Admin page in the sidebar.

> The Admin page is only visible to users with superuser privileges. If you need admin access, ask an existing superuser to upgrade your account.

Q: I forgot my password.

> Contact your system administrator or a superuser to reset your password through the Admin panel.

Q: The chart looks empty or shows very few data points.

> Try changing the time range. If you selected "1D" but there's no intraday data, the chart may appear empty. Switch to "1W", "1M", or "ALL" to see more data.

Q: Can I export predictions or chart data?

> This feature is not currently available in version 1.0.0. It is planned for a future release.

12. Glossary

Term	Definition
OHLCV	Open, High, Low, Close, Volume - the five standard data points recorded for each trading session
Candlestick Chart	A chart type where each bar shows the open, high, low, and close prices for a trading period. Green = price went up, Red = price went down.
Line Chart	A chart that connects closing prices with a continuous line, showing the overall trend
RSI	Relative Strength Index - a momentum indicator measuring the speed and magnitude of price changes. Values above 70 suggest overbought, below 30 suggest oversold.
MACD	Moving Average Convergence Divergence - a trend-following indicator that shows the relationship between two moving averages of a stock's price
Bollinger Bands	A volatility indicator consisting of a moving average with upper and lower bands set at standard deviations away
XGBoost	Extreme Gradient Boosting - the machine learning algorithm used by MarketSight to generate price predictions
Confidence Score	A value from 0% to 100% indicating how reliable the model considers its prediction, based on market conditions
Prediction Overlay	A visual element on the chart (orange dashed line) showing where the model predicts the price will go
Ticker Symbol	A short abbreviation used to identify a publicly traded stock (e.g., AAPL for Apple Inc.)
Sector	A broad category grouping companies by industry (e.g., Technology, Healthcare, Energy)
Volume	The number of shares traded during a given period. Higher volume indicates more market activity.
Volatility	A measure of how much a stock's price fluctuates. High volatility means larger price swings.
Superuser	An account with administrative privileges, including the ability to manage other user accounts
Airflow	An automated pipeline system that handles data loading and model retraining on a schedule
Dashboard	The main overview page showing summary statistics about the system
MVP	Minimum Viable Product - the first functional version of the software with core features

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